

scripts/autonomous-qa/record-qa-session.sh — User Guide

Last verified: 2026-07-26 (LVA-018 script-docs backfill cycle)
Inheritance: HelixConstitution §11.4.18 (script docs); Lava §6.J / §6.Z (anti-bluff evidence) **Classification:** project-specific

Overview

Records the Android emulator screen during an autonomous QA run and saves a **verified** MP4 to ~/Downloads/lava-qa-<timestamp>-<scenario>.mp4.

The recorder uses `adb shell screenrecord` in the background. Because `screenrecord` is capped at 180 seconds per invocation, long tests are recorded as sequential chunks that are concatenated with `ffmpeg` after the test command exits. Chunks are staged under `/tmp` so large captures never bloat the repo.

Anti-bluff contract (§6.J / §6.Z)

The script **refuses to claim success** unless the recorded file is non-empty AND `ffmpeg` confirms it contains a video stream with at least one decoded frame. An empty/corrupt/unprobed recording is exit 2, never a silent "recorded".

Usage

```
scripts/autonomous-qa/record-qa-session.sh --serial <serial> --
scenario <name> [options]
```

Flag	Required	Default	Meaning
--serial	yes	—	adb serial of the running emulator/device
<serial>	yes	—	

Flag	Required	Default	Meaning
--scenario <name>			short scenario slug for the filename (normalized to lowercase alphanumerics + hyphen)
--test-command <cmd>	no	sleep 5	shell command to run while recording
--output-dir <dir>	no	~/Downloads	directory for the final MP4
--bit-rate <bps>	no	4000000	screenrecord bit-rate
--chunk-sec <secs>	no	180	chunk length; MUST be ≤ 180 (screenrecord limit)
-h --help	no	—	show help

Environment: ANDROID_HOME (adb location), FFMPEG (ffmpeg path override).

Exit codes

- 0 — test command passed AND video verified
- 1 — test command failed BUT video was verified (the recording is still valid evidence of the failure)
- 2 — setup error (missing args, adb/ffmpeg unavailable, device unreachable) or video verification failed

Examples

```
# Record a Challenge run end-to-end
scripts/autonomous-qa/record-qa-session.sh \
  --serial 127.0.0.1:42317 --scenario search-flow \
  --test-command './gradlew :app:connectedDebugAndroidTest'
  -Pandroid.testInstrumentationRunnerArguments.class=java.app.challenges.Challenge

# Quick 5-second smoke recording
scripts/autonomous-qa/record-qa-session.sh --serial emulator-5554
  --scenario smoke
```

Companion files

- `scripts/autonomous-qa/run-iteration.sh` — embeds the same chunked-recording pattern inline for matrix iterations
- `scripts/autonomous-qa/vision-analyze.sh` — analyzes the recordings this produces
- `scripts/record-challenge-video.sh` / `scripts/record-device-session.sh` (+ their docs) — the sibling recording paths